Digital UI/UX Timetable

A Project Report

23AID215 - User Interface Design

Submitted By,

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Introduction

The Digital Classroom Timetable is a web-based scheduling application This project aims to replace traditional paper-based or static digital timetables with a fully interactive, dynamic interface that can be managed in real time.

In modern educational institutions, managing class schedules is a time-consuming and error-prone process. Faculty members often need to update schedules manually, and students rely on static notice boards or printed sheets to track their weekly timetable. This project addresses these challenges by providing a single, unified platform where schedules can be viewed, added, edited, and reorganised with ease.

The application is built entirely using HTML, CSS, and JavaScript — three front-end technologies with no backend. The interface supports two distinct roles: a Student View for read-only schedule browsing, and a Faculty View with full editing capabilities. Key features include colour-coded subject blocks, online class indicators with join links, a class swap mechanism, and a Monthly Report page with attendance and statistics.

The project demonstrates core UI/UX principles, including role-based access, intuitive navigation, responsive design, and real-time data updates within a single-page application architecture.

Individual Contribution

CB.AI.U4AID24111(C MANASA) - Timetable Page Structure (HTML)

Responsible for designing and building the core HTML structure of the timetable. This includes the overall page layout, navigation tabs, the role switcher(Student/Faculty), the main timetable, a table with all period rows, break rows, exam rows, and the subject legend.Also structured the modal dialog used for adding and editing classes.

CB.AI.U4AID24160(VARSHINI B) - Styling and Visual Design

Handled the complete visual design of the application using CSS. This includes the colour-coded subject themes(8 colour classes), the responsive timetable layout, the modal popup styling, stat cards on the report page, the swap animation, the legend section and all hover and transition effects.

CB.AI.U4AID24145(RAJARAJESHWARI S) - Interactivity and Logic

Developed the JavaScript that powers the application. This includes the role-switching logic (students vs faculty), the cell-click handler, the add/edit/delete modal flow, the colour selector, the class-swap mode with two-step selection, online link toggling, and keyboard support. Implemented dynamic statistics calculation for the report page.

CB.AI.U4AID24154(SNEHA AS)- Monthly Report Page

Responsible for designing and building the Monthly Report Page. This includes the four statistics cards(Total classes, Online classes, Average Attendance, Exams Conducted), the subject-wise class distribution grid, the Faculty Teaching hours section and the class-wise attendance Rate table with colour- coded status badges.

Problem and User Description:

Problem statement:

- Printed timetables cannot be updated immediately when a class is rescheduled or cancelled.
- Faculty members have no easy way to swap or reassign classes.
- Online and offline classes are not distinguishable in existing timetables, confusing for students.
- Students and faculty use the same interface, creating a need for role-based access to prevent accidental edits.

The Digital Classroom Timetable solves all of these problems by providing a live, role-aware, editable timetable with reporting capabilities.

User Description:

The application is designed for two primary user types:

- Students: Students use the application in read-only mode to view their weekly class schedule. They can see subject names, teacher names, room numbers, online/offline status, and join links for virtual classes. The student view prevents accidental modification of the timetable.
- Teacher: Faculty members can switch to Faculty view to manage the timetable. They can add new classes, edit existing ones, delete entries, swap two classes, and mark classes as online with meeting links. Faculty also have access to the Monthly report for tracking class statistics and attendance data.

UI Design Explanation

Layout:

The application follows a single-page layout with two main sections accessible via navigation tabs at the top: the Timetable page and the Monthly Report page. The overall container is centered with a maximum width of 1400px, giving it a clean, card-like appearance against a light gray background. All content is contained within a white card with rounded corners and a subtle drop shadow.

Color Scheme:

The design uses a deliberate two-tone primary palette of green and red, with white as the base background:

- Green — Used for the h1 title, timetable column headers, all buttons (default state), navigation tab text, week-info text, faculty hours text, and attendance table headers.
- Red — Used for the active navigation tab underline and text, modal header title, form focus border, and the delete button.
- Blue — Used for button hover state and the active role button (Faculty View).
- Grey — Used for subtitle text, form labels, teacher/room info, legend text, and inactive role button background.
- Cyan — Used as the swap info banner background to clearly signal swap mode is active.

Each subject is assigned a distinct named color for instant visual identification in the timetable:

- Mathematics — Pink
- English — LightGreen
- Physics — Violet
- Chemistry — Teal
- Art & Design — SeaGreen
- History — Tomato
- Computer Science — Red
- Physical Education — Orange

Break cells use a LightYellow background with orange text. Exam cells use a LightBlue background with white text and a grey border.

Navigation:

The top navigation uses a flex row with a 30px gap between tabs (no bottom border line, unlike the original). Inactive tabs are shown in green text. The active tab switches to red text with a red bottom border underline, giving a clear visual indication of the current page.

Role Switcher:

The top navigation uses a flex row with a 30px gap between tabs (no bottom border line, unlike the original). Inactive tabs are shown in green text. The active tab switches to red text with a red bottom border underline, giving a clear visual indication of the current page.

Components Used:

- Timetable grid — HTML table with 7 period rows, 2 break rows, 1 exam row, and 5 day columns. Headers are solid green with white text.
- Class cells — Colored blocks using named CSS colors showing subject, teacher, room, and optional join link with a white border.
- Modal dialog — White overlay form with a red title header, grey labels, red focus border on inputs, a blue-bordered selected color option, and a red delete button.
- Swap info banner — Cyan background banner guiding the user through the two-step swap process.
- Legend — White background flex row with 30x20px color swatches and grey labels for each subject.
- Stats cards — Solid-color cards (Indigo, Green, Blue, Teal) on the report page showing key metrics in white text.
- Attendance table — Green header, alternating LightGrey rows, with colored text badges for attendance status.

Individual Pages

Page 1 — Timetable Page (HTML Structure):

The Timetable page is the main page of the application. It contains the navigation tabs, role switcher, swap info banner, controls bar, the timetable table, and the subject legend. The HTML is structured semantically with a container div, a nav-tabs div, role-switcher buttons, and an HTML table element with thead and tbody. Each cell in the tbody is a td with the class 'class-cell'. Filled cells carry data attributes (data-subject, data-teacher, data-room, data-color, data-online, data-link) which are used by JavaScript to read and write class information.

Page 2 — CSS Styling:

The CSS file (styles.css) provides all visual styling. The body and container both use a white background. Green is the primary theme color applied to headings, table headers, buttons, and nav tabs. Red is the active/accent color for tab states, modal headers, and focus rings. The 8 subject color classes use named CSS colors (Pink, LightGreen, Violet, Teal, SeaGreen, Tomato, Red, Orange) applied as background to filled class cells. The swap selected state uses a green border and a LightGrey background. Role buttons have border-radius: 10px (rounded), compared to the standard 5px used elsewhere. The legend color swatches are 30x20px. Media queries at 768px reduce padding and font sizes for mobile.

Page 3 — JavaScript Logic:

The script.js file handles all interactive behavior. The switchRole() function toggles between student and faculty modes by adding/removing CSS classes and showing/hiding controls. The handleCellClick() function routes clicks to either editCell() or handleSwap(), depending on the current mode. The modal functions (openAddModal, editCell, closeModal, saveSubject, deleteSubject) manage the form lifecycle. The handleSwap() function implements a two-click swap: the first click selects a cell, the second performs the data exchange between two cells. The updateStatistics() function counts filled cells and online cells in real time and updates the report page figures.

Page 4 — Monthly Report Page:

The Monthly Report page displays a timetable data summary. Four solid-color stat cards use Indigo, Green, Blue, and Teal backgrounds with white text. Below the cards, a subject breakdown grid shows per-subject class counts in white cards with a green border (border: 1px solid green). The faculty teaching hours section shows teacher cards with a grey name, grey subjects, and green hours text. The attendance table uses a green header row, alternating LightGrey even rows, and colored text badges — green text for high attendance, orange for medium, and red for low. The h3 titles in the report section have a green bottom border.

Code Samples:

HTML part:

Timetable Cell structure

```
<!-- Period 2: 9:40 - 10:30 AM -->
<tr>
    <td class="time-column">9:40 - 10:30 AM</td>
    <td class="class-cell filled color-2" onclick="handleCellClick(this)" data-subject="English" data-t
        <div class="subject-name">English</div>
        <div class="teacher-name">Ms. Emma Davis</div>
        <div class="room-number">Room 315</div>
    </td>
    <td class="class-cell filled color-4" onclick="handleCellClick(this)" data-subject="Chemistry" data
        <div class="subject-name">Chemistry <span class="online-badge">📹</span></div>
        <div class="teacher-name">Dr. Michael Brown</div>
        <div class="room-number">Lab 202</div>
        <a href="https://zoom.us/j/123456789" class="online-link" target="_blank" onclick="event.stopPr
    </td>
    <td class="class-cell filled color-7" onclick="handleCellClick(this)" data-subject="Computer Scienc
        <div class="subject-name">Computer Science</div>
        <div class="teacher-name">Ms. Lisa Park</div>
        <div class="room-number">Lab 305</div>
    </td>
    <td class="class-cell filled color-4" onclick="handleCellClick(this)" data-subject="Chemistry" data
        <div class="subject-name">Chemistry</div>
        <div class="teacher-name">Dr. Michael Brown</div>
        <div class="room-number">Lab 202</div>
    </td>
    <td class="class-cell filled color-1" onclick="handleCellClick(this)" data-subject="Mathematics" da
        <div class="subject-name">Mathematics</div>
        <div class="teacher-name">Dr. Sarah Chen</div>
        <div class="room-number">Room 204</div>
    </td>
</tr>
```

Navigation tabs

```
<!-- Navigation Tabs -->
<div class="nav-tabs">
    <button class="nav-tab active" onclick="switchPage('timetable')">Timetable</button>
    <button class="nav-tab" onclick="switchPage('report')">Monthly Report</button>
</div>
```

CSS part :

Named CSS colors used for subject cells and online link

```
.online-link {
    display: inline-block;
    margin-top: 4px;
    padding: 2px 6px;
    border-radius: 3px;
    font-size: 10px;
    text-decoration: none;
    color: white;
    border: 1px solid white;
}


.online-badge {
    display: inline-block;
    padding: 1px 4px;
    background:white;
    border-radius: 3px;
    font-size: 9px;
    margin-left: 3px;
}

/* Color themes for subjects */
.color-1 { background: Pink;}
.color-2 { background: LightGreen ;}
.color-3 { background: Violet;}
.color-4 { background: Teal; }
.color-5 { background: SeaGreen; }
.color-6 { background: Tomato; }
.color-7 { background: Red; }
.color-8 { background: Orange; }
```

Attendance table

```
.attendance-table th,
.attendance-table td {
    padding: 12px;
    text-align: left;
    border: 1px solid LightGray;
}

.attendance-table th {
    background: Green;
    color: white;
    font-weight: bold;
}

.attendance-table tr:nth-child(even) {
    background: LightGrey;
}

.attendance-rate {
    display: inline-block;
    padding: 4px 10px;
    border-radius: 4px;
    font-weight: bold;
    font-size: 12px;
}

.attendance-rate.high {
    color: green;
}
```

JavaScript

Switch between Faculty view and student view

```
function switchRole(role) {
    currentRole = role;

    // Update button states
    document.querySelectorAll('.role-btn').forEach(btn => {
        btn.classList.remove('active');
    });
    event.target.classList.add('active');

    // Show/hide faculty controls
    const facultyControls = document.getElementById('faculty-controls');
    const timetable = document.getElementById('timetable');

    if (role === 'faculty') {
        facultyControls.classList.add('active');
        timetable.classList.remove('student-view');
    } else {
        facultyControls.classList.remove('active');
        timetable.classList.add('student-view');
        // Exit swap mode if active
        if (swapMode) {
            toggleSwapMode();
        }
    }
}
```

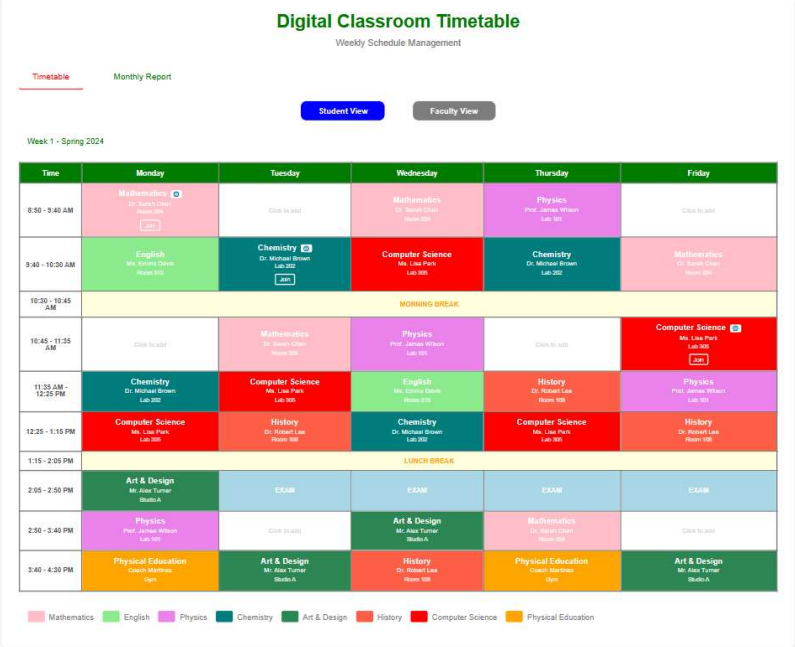
Swapping the classes

```
function handleSwap(cell) {
    if (!cell.classList.contains('filled')) {
        alert('Cannot swap with empty cell');
        return;
    }

    if (!firstSwapCell) {
        // Select first cell
        firstSwapCell = cell;
        cell.classList.add('selected-for-swap');
        document.getElementById('swap-message').textContent = 'Now click on another class to swap with';
    } else {
        // Perform swap
        if (firstSwapCell === cell) {
            // Clicked same cell, deselect
            firstSwapCell.classList.remove('selected-for-swap');
            firstSwapCell = null;
            document.getElementById('swap-message').textContent = 'Click on a class to select it, then click another class to swap';
            return;
        }
```

Output Screenshots:

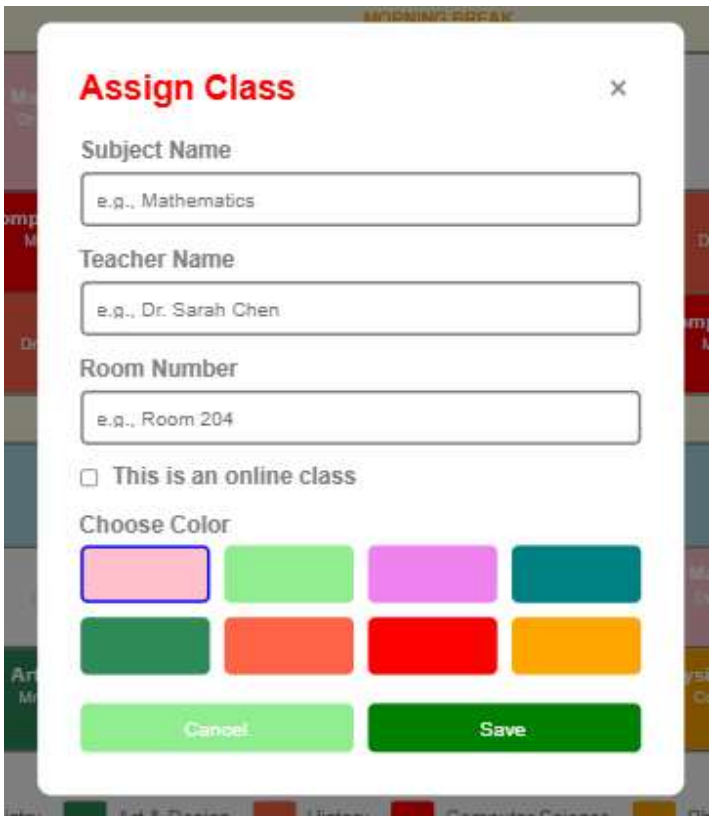
Student View :

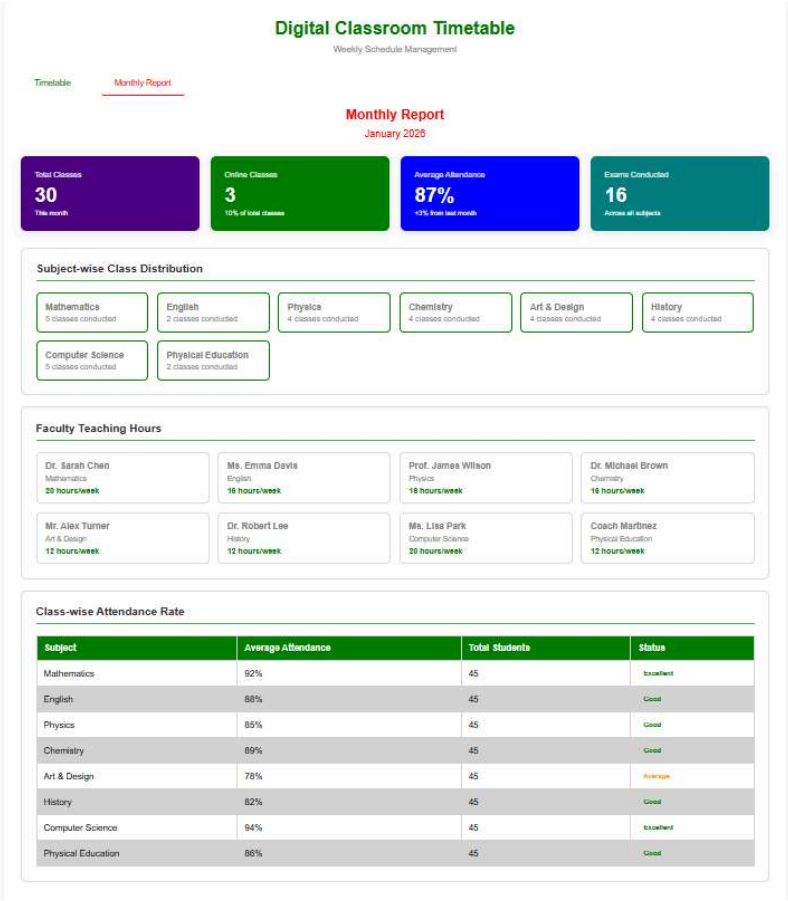


Faculty View:

- Can assign and swap classes

Assign class



Swapping classes

Monthly Report Page